Initial Project Idea

Shaunak Kapur, sk62288 ECE 382V Software Testing in the Era of Nondeterminism, Fall 2026

Are LLM Written Unit Tests Flaky?

For my project I want to study whether unit tests written by large language models are flaky. A lot of developers now use tools like GitHub Copilot or ChatGPT to write their tests and research tools like TestPilot generate a test and run it one time and keep it if it passes. The problem is that a flaky test can pass that one run and then fail later in CI. Also every big flaky test study I have read so far only looks at tests written by people. My plan is to take 12 small Python libraries and give each library four test suites. These are the human written suite that comes with the library and a suite from an agent that runs its tests and fixes the failing ones and two one shot suites that are never run while they are written, one from Claude Sonnet and one from Claude Haiku. I rerun every suite 70 times while changing the test order and the hash seed and the time zone and the CPU load, and I mark a test as flaky if it both passes and fails across those runs. I already have a first version of this working with about 714,000 test executions. So far 6 of the 5,587 human tests are flaky and 4 of those are order dependent. In addition 3 of the 1,595 one shot Haiku tests are flaky and all 3 are about real time. All 3 of those tests also passed the one run filter, which means a real pipeline would have kept them. Furthermore the LLM suites have about twice as many tests with risky patterns like real sleeps and clock reads that are not frozen. To evaluate this I will report the flaky rate of each suite with confidence intervals and the root cause of every flaky test using the categories from Luo et al. I will also check how well the risky patterns predict real flips. Later I want to try a prompt that warns the model about these patterns and a filter that reruns each test a few times before keeping it. However the part I am most unsure about is that flaky tests are rare, so I may need a lot more reruns or a slower machine like a Colab VM to see a real difference between the suites. I am working on this project individually.

Team

I am working on this project individually.

Lightning talk slides

The 5 slides for the 5 minute lightning talk are on the following pages. The editable PowerPoint version with speaker notes is Kapur_ECE382V_Initial_Idea.pptx.

Are LLM Written Unit Tests Flaky?

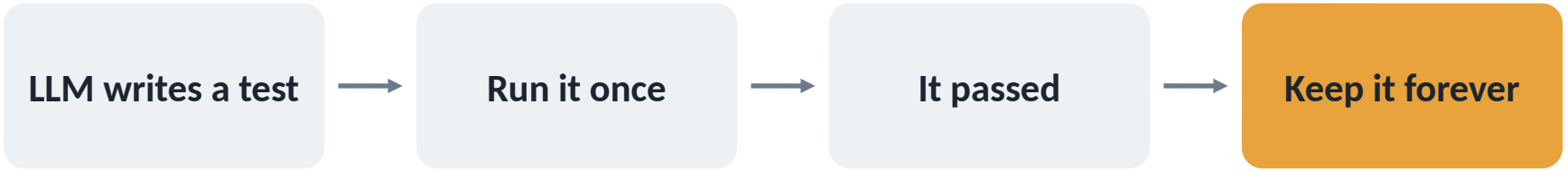
Initial project idea for ECE 382V

Shaunak Kapur, working individually



30 reruns of one test, one of them failed

Generation pipelines trust one passing run



- TestPilot and most Copilot style pipelines keep a generated test if it passes one run.
- Every big flaky test study so far looks at tests written by people.
- So I want to know if LLM tests are more flaky, why, and if one run is enough to catch it.



A flaky test can pass the one run that decides if it gets kept.

What I am running

12 Python libraries

cachetools	schedule	tinydb
humanize	toolz	tabulate
tenacity	filelock	boltons
more-itertools	arrow	pybreaker

Tan ones were added because they use time, randomness, threads, or files.

4 test suites per library

Human
the library's own test suite

Agentic Sonnet
runs its tests and fixes failures

One shot Sonnet
writes tests, never runs them

One shot Haiku
same, with a smaller model

Every suite reruns 70 times with

- random order and hash seed
- 10 time zones
- CPU stress on every core

Early results

714,000

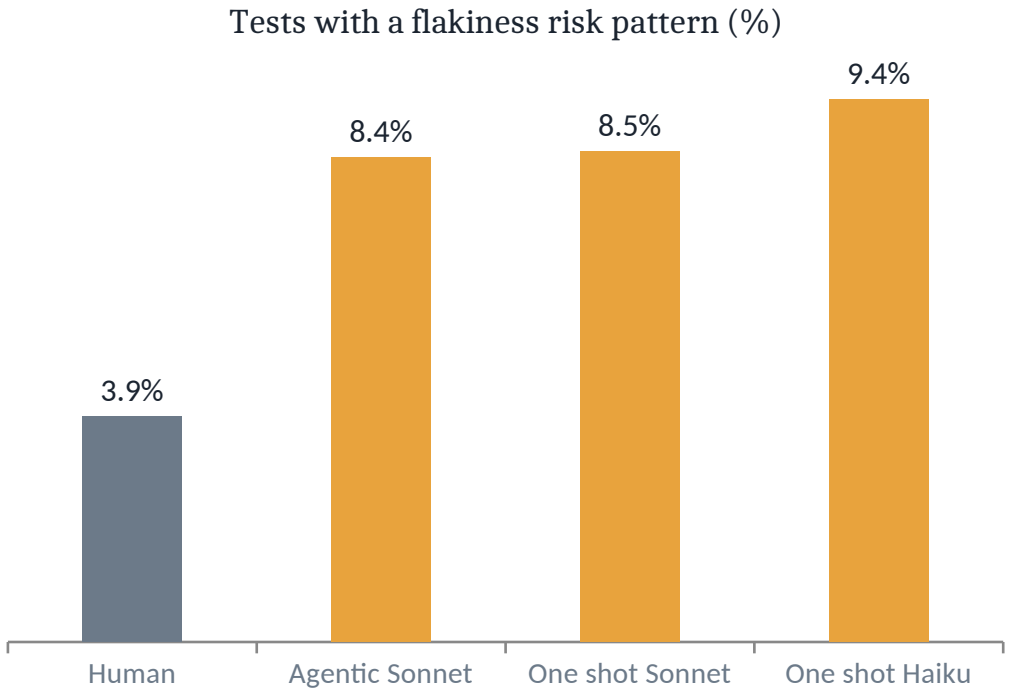
test executions across 48 suites

3

LLM tests flipped, all about real time, and a one run filter kept all of them

6

human tests flipped, mostly because of test order



Next steps and open questions

How I will evaluate

- Flaky rate per suite with confidence intervals
- Root cause of every flaky test
- Risk patterns and coverage per suite
- What a one run filter keeps and throws away

Timeline

- Oct 9 proposal
- Oct 23 more models and reruns
- Nov 6 slower CI machine runs
- Nov 20 mitigation prompt
- Dec 12 final report

What I am unsure about

- Flips are rare, so do I need 1000 reruns
- Do risk patterns predict real flips
- Is a laptop too fast to show timing bugs
- Ideas for other perturbations welcome